Mandatory assignment 1.

1.2.3 - Exact solution

$$u(x,y,t) = e^{i(k_x x + k_y y - \omega t)}$$

. Laplacian operator ∇^2 in 2D : $\nabla^2 u = u_{xx} + u_{yy}$

. $u_t = -i\omega e^{i(k_x x + k_y y - \omega t)} = -i\omega u$

. $u_{tt} = (-i\omega)^2 u = -\omega^2 u$

. $u_x = ik_x u$; $u_{xx} = (ik_x)^2 u = -k_x^2 u$ and $u_y = ik_y u$; $u_{yy} = (ik_y)^2 u = -k_y^2 u$

$$\Rightarrow u_{xx} + u_{yy} = -(k_x^2 + k_y^2) u$$

we know : $u_{tt} = c^2 \nabla^2 u = c^2 (u_{xx} + u_{yy}) \Rightarrow -\omega^2 u = c^2 [-(k_x^2 + k_y^2) u]$

$$\Rightarrow \omega^2 = c^2 [-(k_x^2 + k_y^2)]$$

$$\Rightarrow \omega = \sqrt{c^2 [-(k_x^2 + k_y^2)]}$$

1.2.4 - Dispersion coefficient

$$C = \frac{c\Delta t}{h}$$

from the standard 2D 2nd order explicit scheme.

(1) $u^{n+1}_{i,j} = 2u^n_{i,j} - u^{n-1}_{i,j} + C^2 [(u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}) + (u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1})]$

and equation 1.7. (2) $u^n_{i,j} = e^{i(kh(i+j) - \tilde{\omega} n \Delta t)} = e^{(i(ik_x + jk_y - n\theta)}$ with $k_x = k_x h$, $k_y = k_y h$, $\theta = \tilde{\omega}\Delta t$

numerical frequency (dispersion)

Time : plug into (1)

$$u^{n+1}_{i,j} - 2u^n_{i,j} + u^{n-1}_{i,j} = u^n_{i,j}(e^{-i\theta} - 2 + e^{i\theta}) = u^n_{i,j}(2\cos\theta - 2) = u^n_{i,j}\left(-4\sin^2\left(\frac{\theta}{2}\right)\right)$$

Space :

x-direction : $u^n_{i+1,j} = e^{i((i+1)k_x + jk_y - n\theta)} = e^{i(ik_x + jk_y - n\theta)} e^{ik_x} = u^n_{i,j} e^{ik_x}$

$$u^n_{i-1,j} = u^n_{i,j} e^{-ik_x}$$

so $(u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}) = u^n_{i,j}(e^{ik_x} - 2 + e^{-ik_x}) = u^n_{i,j}(2\cos k_x - 2) = u^n_{i,j}\left(-4\sin^2\left(\frac{k_x}{2}\right)\right)$ (3)

y-direction : same, $(u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1}) = u^n_{i,j}\left(-4\sin^2\frac{k_y}{2}\right)$ (4)

(3) + (4) $\Rightarrow [(u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}) + (u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1})] = -4\left[\sin^2\left(\frac{k_x}{2}\right) + \sin^2\left(\frac{k_y}{2}\right)\right] u^n_{i,j}$

$$\Rightarrow \sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = C^2\left[\sin^2\left(\frac{k_x h}{2}\right) + \sin^2\left(\frac{k_y h}{2}\right)\right]$$

$$k_x = k_y = k$$

$$\Rightarrow \sqrt{k_x^2 \cdot k_y^2} = \sqrt{2}\,k$$

$$\Rightarrow \sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = 2C^2\sin^2\left(\frac{kh}{2}\right) = \sin^2\left(\frac{kh}{2}\right) \Rightarrow \tilde{\omega}\Delta t = kh$$

$$\Rightarrow \tilde{\omega} = \frac{2}{\Delta t}\sin^{-1}\left(C\sqrt{\sin^2\left(\frac{k_x h}{2}\right) + \sin^2\left(\frac{k_y h}{2}\right)}\right) \quad \text{with } C \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow \tilde{\omega} = \frac{kh}{\Delta t} = k\frac{c}{C} = k\left(\sqrt{2}c\right) = c\sqrt{k_x^2 + k_y^2} = \omega$$

$$\Rightarrow \text{so } \tilde{\omega} = \omega$$